

Machine Learning

Machine learning (ML) is a subset of artificial intelligence (AI) that enables computers to learn from data and make predictions without being explicitly programmed. Instead of following hard-coded rules, ML algorithms identify patterns in training data to build models that can generalize to new, unseen information. [1, 2, 3]

How It Works

Machine learning operates on a core premise: feed an algorithm a massive amount of data, and the model will "learn" the underlying rules of that information. The quality of the outcome directly correlates to the quality and relevance of the data fed into it (often referred to as "garbage in, garbage out")

The Core Types of Machine Learning

- **Supervised Learning:** The model is trained on labeled data (where both the input and the correct output are known). It learns to map inputs to outputs for tasks like *classification* (e.g., detecting spam) or *regression* (e.g., predicting house prices). [1]
- **Unsupervised Learning:** The model is given unlabeled data and must find hidden patterns or structures on its own. Common uses include *clustering* (grouping customers by purchase history) or *dimensionality reduction*. [1, 2, 3, 4]
- **Reinforcement Learning:** An AI "agent" learns to make decisions by taking actions in an environment to maximize some notion of cumulative reward. It learns through trial and error, much like how a human learns to play a video
- **Generative AI:** A rapidly growing subset that learns the patterns and structures of input data to generate new, original content (like text, images, or code)

Real-World Applications

Machine learning powers many of the tools and services you interact with daily: [1]

- **Recommendation Systems:** Platforms like Netflix or YouTube analyzing what you watch to suggest future content.
- **Computer Vision:** Self-driving cars identifying pedestrians or manufacturing machines inspecting products for defects.
- **Healthcare:** Algorithms scanning medical imagery to assist doctors in diagnosing conditions.

- **Fraud Detection:** Banks analyzing transaction patterns in real-time to flag suspicious activity. [[1](#), [2](#), [3](#), [4](#), [5](#)]

Getting Started

For those looking to get into the field, the modern ML landscape relies heavily on the **Python** programming language, supported by foundational libraries like NumPy (for arrays and math), Pandas (for data manipulation), and Scikit-Learn or PyTorch (for building models). [[1](#), [2](#), [3](#), [4](#), [5](#)]

For a beginner-friendly, high-level breakdown of how machines learn to recognize patterns in data without explicit instructions:

Data Collection

Collecting data is the most important part of any Machine Intelligence projects.

The most common data to collect are numbers and measurements.

Often data are stored in arrays representing the relationship between values.

This table contains house prices versus size:

Frank Rosenblatt

Frank Rosenblatt (1928 – 1971) was an American psychologist notable in the field of Artificial Intelligence.

In **1957** he started something really big. He "invented" a **Perceptron** program, on an IBM 704 computer at Cornell Aeronautical Laboratory.

Scientists had discovered that brain cells (**Neurons**) receive input from our senses by electrical signals.

The Neurons, then again, use electrical signals to store information, and to make decisions based on previous input.

Frank had the idea that **Perceptrons** could simulate brain principles, with the ability to learn and make decisions.

The Perceptron

The original **Perceptron** was designed to take a number of **binary** inputs, and produce one **binary** output (0 or 1).

The idea was to use different **weights** to represent the importance of each **input**, and that the sum of the values should be greater than a **threshold** value before making a decision like **yes** or **no** (true or false) (0 or 1).

Perceptron Example

Imagine a perceptron (in your brain).

The perceptron tries to decide if you should go to a concert.

Is the artist good? Is the weather good?

What weights should these facts have?

Neural Networks are used in applications like Facial Recognition.

These applications use **Pattern Recognition**.

This type of **Classification** can be done with a **Perceptron**.

Perceptrons can be used to classify data into two parts.

Perceptrons are also known as a **Linear Binary Classifiers**

Pattern Classification

Imagine a straight line (a linear graph) in a space with scattered x y points.

How can you classify the points over and under the line?

Create a Perceptron Object

Create a Perceptron object. Name it anything (like Perceptron).

Let the perceptron accept two parameters:

1. The number of inputs (no)
2. The learning rate (learningRate).

Set the default learning rate to 0.00001.

Then create random weights between -1 and 1 for each input.